

AKASH SONI

Backend Engineer | Java · Spring Boot · Microservices · Kafka · AWS · Distributed Systems
Bengaluru, Karnataka | sameersoni396@gmail.com | +91 9685406249 | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Results-driven Backend Engineer with 3.4+ years of experience designing scalable microservices, distributed backend systems, and cloud-native applications using Java, Spring Boot, Kafka, Redis, and AWS. Proven expertise in building high-performance RESTful APIs, event-driven architectures, secure enterprise applications, and AI-powered platforms across healthcare and SaaS domains. Strong focus on low latency, reliability, observability, and production-grade engineering practices.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend & Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Hibernate, FastAPI, REST APIs, JWT Authentication, Microservices, Event-Driven Architecture

Distributed Systems & Messaging: Apache Kafka, Redis, Asynchronous Processing, Caching Strategies, Pub/Sub Messaging

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, API Gateway), Docker, Kubernetes (Basics), Jenkins, CI/CD Pipelines, SonarQube, GitHub Actions

Databases: MySQL, PostgreSQL

Testing & Quality: JUnit, Mockito, TestNG, JMeter, Test-Driven Development (TDD)

Frontend & UI: React, TypeScript, Vite, Tailwind CSS

Observability: Logging & Monitoring, Application Debugging, Performance Tuning, API Monitoring

Tools & Platforms: Git, GitHub, Postman, Swagger/OpenAPI, Jira, VS Code, Eclipse IDE

Core Concepts: Distributed Systems, System Design, Design Patterns, Scalability Engineering, API Security, Performance Optimization, Agile & Scrum, Low-Latency & High Availability Systems

PROFESSIONAL EXPERIENCE

Tata Elxsi Ltd. — Software Developer (Backend Engineering)

Jan 2023 – Present

Bengaluru, Karnataka

- Architected scalable microservices-based backend systems using Java and Spring Boot, improving service modularity and reducing cross-service dependencies by 40%.
- Engineered high-throughput RESTful APIs and API gateway integrations, increasing system throughput by 30% while reducing API latency by 25% under peak load conditions.
- Optimized backend application performance by 45% through asynchronous processing, Kafka-based event pipelines, query optimization, and legacy service refactoring.
- Implemented caching strategies and optimized database interactions for low-latency API responses and improved backend scalability.
- Developed secure authentication and authorization workflows using JWT and Spring Security for enterprise-grade healthcare platforms.
- Increased automated test coverage from below 50% to 77% using JUnit, Mockito, and TestNG, reducing production defects by 20%.
- Collaborated with DevOps teams to implement CI/CD pipelines using Jenkins, Docker, and GitHub Actions, reducing deployment failures by 30%.
- Integrated third-party healthcare APIs and real-time data services enabling seamless interoperability across distributed healthcare ecosystems.
- Participated in production support, debugging, performance tuning, and monitoring for scalable distributed systems in enterprise environments.
- Contributed to Agile/Scrum cycles including sprint planning, peer code reviews, release management, and feature delivery.

PROJECTS

IDP Intel Engine — Intelligent Document Processing Platform

FastAPI · Python · React · MySQL · Google Gemini AI · JWT · Docker

- Built a multi-tenant SaaS Intelligent Document Processing platform for automated extraction, analysis, and digitization of medical and insurance documents using Google Gemini AI.

- Designed scalable FastAPI microservices with JWT-based RBAC authentication, enabling secure multi-organization data isolation and workflow management.
- Developed AI-powered extraction pipelines with Human-in-the-Loop (HITL) validation workflows, improving document processing accuracy and operational efficiency.
- Implemented asynchronous document processing, metadata management, and high-fidelity PDF digitization for scalable enterprise document workflows.

MultiModel-LLM — Unified Multi-Provider AI Platform

Java · Spring Boot · React · TypeScript · MySQL · Redis · JWT · AES-256

- Developed a secure multi-model AI platform integrating OpenAI, Gemini, Claude, and Llama models through a unified high-performance interface.
- Built scalable Spring Boot backend services with reactive API integrations, Redis-based caching, and standardized orchestration for low-latency LLM interactions.
- Implemented encrypted API key vault storage using AES-256 encryption with secure key management and rotation workflows.
- Designed React + TypeScript frontend supporting dynamic model switching and optimized AI interaction workflows.

Label Process — AI-Powered MDR Compliance Audit System

FastAPI · Python · React · MySQL · Google Gemini · Tesseract OCR

- Developed an AI-powered compliance auditing platform for medical device labels aligned with EU MDR 2017/745 regulations.
- Implemented multimodal AI workflows combining OCR extraction and Gemini-based image analysis to detect missing ISO 15223-1 symbols and compliance gaps.
- Built backend services for annotation processing, audit tracking, and structured compliance reporting with real-time monitoring dashboards.

FHIR R4 Healthcare Interoperability Platform

Java · Spring Boot · PostgreSQL · Kafka · Microservices

- Developed FHIR R4-compliant healthcare integration services enabling secure interoperability between hospitals, insurers, and healthcare applications.
- Designed scalable REST APIs and asynchronous processing pipelines for secure transmission of patient vitals, medical history, and device data.
- Integrated EHR/EMR systems and third-party healthcare APIs to automate care coordination, remote monitoring, and patient risk analysis workflows.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Confluent Certified Data Streaming Engineer (Kafka)
- REST API Intermediate — HackerRank
- SQL Advanced — HackerRank
- AWS Essential Training for Developers — LinkedIn Learning
- Programming Foundations: APIs and Web Services — LinkedIn Learning

HONORS & AWARDS

- Ownership Award — Tata Elxsi
- Project Excellence Award — Tata Elxsi
- Extra Mile Award — Recognized for outstanding backend ownership, technical delivery, and cross-team contribution

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior

2018 – 2022

Bachelor of Technology (B.Tech) — Information Technology | CGPA: 7.77